

Lab 01 — Lab Tasks

Classes and Objects: Fundamentals

Attempt each task on your own using the hint provided. Work through them in order — Easy, then Medium, then Hard.

Easy Level

Task 1: Student Class with Getters and Setters

Create a class Student with private data members name (string) and marks (int). Provide public setter and getter functions for both. In main(), create an object, set its values using the setters, then print them using the getters.

Task 2: Circle Class with a const Getter

Create a class Circle with a private double radius, a public setRadius() setter, and a public const member function getArea() that returns $\pi \times \text{radius}^2$. Create an object, set the radius, and print the area.

Medium Level

Task 3: Out-of-Line Functions with a Private Helper

Create a class Circle where calculateArea() and calculatePerimeter() are declared inside the class but defined outside it using the scope resolution operator. Add a private member function validateRadius() that returns true only if radius > 0, and have your setRadius() call it before assigning.

Task 4: Object, Pointer, and Reference Together

Create a class Employee with private members and public getters/setters. In main(), create one object normally, one dynamically using new and a pointer, and one reference that aliases the first object. Access and print members through all three: dot notation, arrow notation, and the reference.

Hard Level

Task 5: BankAccount with an Internal Validation Helper

Create a BankAccount class with a private double balance, a private member function isValidAmount(double) that checks the amount is positive, and public deposit()/withdraw() functions that call isValidAmount() internally before changing balance. Add a public const getBalance(). Test with both valid and invalid amounts.

Task 6: Discount Function Using Pointers into an Array of Objects

Create a Product class with private members name and price (with public getters/setters). Write a non-member function void applyDiscount(Product* p, double percent) that reduces the pointed-to object's price by the given percentage. In main(), create an array of 3 Product objects, initialize them, apply a discount to each using their address, then print all three in a loop.